

The attrtoolbox package

To be V.0.1

<https://github.com/jandermoreira/attrtoolbox>

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2025/04/27

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ChangeHistory

0.1 (2025/04/27)
Initial version.

1 Introduction

This is a general package to handle keys and values in an easy way. One may think this as declaring variables in a programming language.

2 Usage

This package can be used by loading it in the preamble.

```
\usepackage[<options>]{attrtoolbox}
```

This is the only available package option.

```
readonly = true | false
```

Initially: false

When set to **true** any attempt to change the value of an attribute will result in error. Otherwise the value can be changed freely.

```
\usepackage[readonly]{attrtoolbox}
```

3 Commands

3.1 Attributes

```
\ATAttributeSet[⟨class⟩]{⟨attribute⟩}{⟨value⟩}
```

This sets an attribute *⟨attribute⟩* to *⟨value⟩*. The *⟨class⟩* is optional.
See `\ATClassSet`.

```
\ATAttributeSetFrom[⟨class⟩]{⟨attribute⟩}{⟨macro⟩}
```

This sets an attribute *⟨attribute⟩* to the contents *⟨macro⟩* (one expansion). The *⟨class⟩* is optional.
See `\ATClassSet`.

```
\ATAttributeGet[⟨class⟩]{⟨attribute⟩}
```

This expands the value of the attribute *⟨attribute⟩* in class *⟨class⟩*.

```
\ATAttributeSet{name}{Ann}
Hi, \ATAttributeGet{name}!

\ATAttributeSet{name}{John} % new content
Hi, \ATAttributeGet{name}!
```

Hi, Ann!
Hi, John!

```
\ATAttributeSet{equation}{$E = mc^2$}
\ATAttributeGetTo{\content}{equation}
\edef\formula{\ATAttributeGet{equation}}
The content: \meaning\content\par
The formula: \meaning\formula\par
The formula expanded: \expandafter\meaning\formula
```

The content: macro:->\$E = mc^2\$
The formula: macro:->\ATAttributeGet {equation}
The formula expanded: \protected macro:->__cmd_start:nNNnnn {om}\ATAttributeGet
\ATAttributeGet code {__cmd_grab_D:w []__cmd_grab_m_1:w }}{equation}

```
\ATAttributeGetTo[⟨class⟩]{⟨macro⟩}{⟨attribute⟩}
```

This copies the content of *⟨attribute⟩* in class *⟨class⟩* to *⟨macro⟩*.

Attributes can be used with classes. When `\ATAttributeGet` is used specifying a class name, it is equivalent to `\ATClassGet`. All the same, a class attribute can be set with `\ATAttributeGet`.

```

\ATClassSet{computer}{
    processor = Intel,
    ram = 8GiB,
    hd = 1TB,
}
Processor: \ATAttributeGet[computer]{processor}\par
Main memory (current): \ATAttributeGet[computer]{ram}\par
\ATAttributeSet[computer]{ram}{16GiB}
Main memory (future upgrade): \ATClassGet{computer}{ram}\par
Storage: \ATAttributeGet[computer]{hd}\par

```

```

Processor: Intel
Main memory (current): 8GiB
Main memory (future upgrade): 16GiB
Storage: 1TB

```

`\ATAttributeIfExist[⟨class⟩]{⟨attribute⟩}{⟨code-if-true⟩}{⟨code-if-false⟩}`

Executes `⟨code-if-true⟩` if the `⟨attribute⟩` exists, otherwise execute `⟨code-if-false⟩`.

```

\ATClassSet{someone}{
    forname = John,
    surname = Doe,
}
Name: \ATClassGet{someone}{forname}\ATAttributeIfExist
↪ [someone]{middlename}{~\ATClassGet{someone}{middlename}}{}
↪ \ATClassGet{someone}{surname}.

\ATClassSet{someone}{middlename = K.}
Name: \ATClassGet{someone}{forname}\ATAttributeIfExist
↪ [someone]{middlename}{~\ATClassGet{someone}{middlename}}{}
↪ \ATClassGet{someone}{surname}.

```

```

Name: John Doe.
Name: John K. Doe.

```

```

\ATAttributeSet{title}{Mr.}
Dear \ATAttributeIfEqual{title}{Mr.}{Sir}{Madam}, \ldots

```

```

Dear Sir, ...

```

3.2 Classes

A class is a collection of attributes and their values.

`\ATClassSet{⟨class⟩}{⟨list-of-attributes⟩}`

This macro is a shorthand to set several attributes in a `⟨class⟩`. The `⟨list-of-attributes⟩` is an `attribute = value` comma-separated list.

This command replaces a sequence of `\ATAttributeSet` calls for a single `⟨class⟩`.

```
\ATClassSetFrom{<class>}{<macro>}
```

This macro is a shorthand to set several attributes in a <class>. The <macro> is expected to hold a comma-separated list.

This command replaces a sequence of `\ATAttributeSet` calls for a single <class>.

```
\ATClassGet{<class>}{<attribute>}
```

This macro expands the value of <attribute> from <class>.

Note that `\ATClassGet{myclass}{myattr}` has the same functionality of `\ATAttributeGet[myclass]{myattr}`. (See `\ATAttributeGet`.)

```
\ATClassSet{person}{
  forname = Susan,
  surname = Smith,
  age = 33,
}
\ATClassGet{person}{forname} \ATClassGet{person}{surname} is
↪ \ATClassGet{person}{age} years old.
```

Susan Smith is 33 years old.

4 Other tools

4.1 Lists

```
\ATListCreate{<name>}
```

New empty list

```
\ATListAppendItems{<name>}{<list-of-items>}
```

Add items to list

```
\ATListAppendItemsFrom{<name>}{<macro>}
```

Get items from macro and add them to list

```
\ATListForEach{<list-name>}{<macro>}{<commands>}
```

foreach <macro> in <list-name> do <commands>

When specified, the <separators> must be a four-token argument with {<sep-between-two>}, {<sep-between-more-than-two>}, {<sep-between-final-two>}, and {<after-last>}. If the list has two items then <sep-between-two> is placed between them; if more than two items are present, then <sep-between-more-than-two> is placed between each item except the last, where <sep-between-final-two> is used. Finally, <after-last> (which is not a separator *per se*) is placed after the last item. All separators are empty by default.

There is a counter for each loop which can be accessed by `\ATCounter`.

```

\ATListCreate{cities}
\ATListAppendItems{cities}{Sao Carlos}
\def\MyCities{Paris, London, Rio de Janeiro}
\ATListAppendItemsFrom{cities}{\MyCities}
\begin{itemize}
  \ATListForEach{cities}{\city}{\item \city}
\end{itemize}

```

-
- Sao Carlos
 - Paris
 - London
 - Rio de Janeiro

```

\ATListCreate{cities}
\ATListAppendItems{cities}{Sao Carlos}
My cities: \ATListForEach{cities}{\city}[{ and }, ], and ]!]{\textit{\city}}

\ATListAppendItems{cities}{Paris}
My cities: \ATListForEach{cities}{\city}[{ and }, ], and ]!]{\textit{\city}}

\ATListAppendItems{cities}{London, Rio de Janeiro}
My cities: \ATListForEach{cities}{\city}[{ and }, ], and ]!]{\textit{\city}}

```

My cities: *Sao Carlos*!
 My cities: *Sao Carlos* and *Paris*!
 My cities: *Sao Carlos, Paris, London, and Rio de Janeiro*!

```
\ATForEach{<macro>}{<csv-list>}[<separators>]{<commands>}
```

for <macro> in <csv-list> do <commands>
 The list is created on the fly then discarded.

```
\ATForEachFrom{<macro>}{<macro-with-list>}[<separators>]{<commands>}
```

for <macro> in content of <macro-with-list> do <commands>.
 The list is created on the fly then discarded.

```

My list: \ATForEach{\fruit}{apple, orange, lemon, peach}[{ and }, ], and
↪ ]{.}]{\fruit}

```

```

My list: \ATForEach{\fruit}{apple, orange, lemon, peach}[{ and }, ], and
↪ ]{.}]{(\ATCounter)~\fruit}

```

My list: apple, orange, lemon, and peach.
 My list: (1) apple, (2) orange, (3) lemon, and (4) peach.

```
\ATListSort{<list-name>}
```

Sorts alphabetically the contents of the list.

```

\ATListCreate{a dolgaim}
\ATListAppendItems{a dolgaim}{ágy, tükör, könyv, szék, asztal, ernyő, ajtó,
↪ hűtő, bőrönd, gyűrű}
\ATListSort{a dolgaim}

```

List of my things in Hungarian: `\ATListForEach{a dolgaim}{\thing}[{ és }{, }{,
→ és }{.}]{\thing}`

List of my things in Hungarian: ajtó, asztal, bőrönd, ernyő, gyűrű, hűtő, könyv, szék, tükör, és
ágy.

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